Point-by-point response to reviewers

20 August 2026

Habitual physical activity and motor cortical neurophysiology:
A systematic review with meta-analyses of transcranial magnetic stimulation measures

Our manuscript was initially submitted to a nonprofit, society-published scientific journal, underwent peer review, and was rejected after the first round. We considered the three reviewers' comments useful and addressable. To avoid wasting the effort invested by the reviewers, an increasingly pressing issue in our publishing system, we decided to address their comments. For two of the reviewers, we have provided our responses without reproducing their comments, out of respect for confidentiality and because our responses provide sufficient context to understand the issues raised. The third reviewer signed his report and gave us permission to share his comments.

In the following point-by-point responses, reviewers' comments are presented in bold, our responses in regular font, excerpts from the manuscript in italics, and all changes made to the manuscript are shown in blue.

REVIEWER 1

Response to Comment 1
We thank Reviewer 1 for the positive evaluation of our work. All comments have been carefully considered and addressed, as detailed below.

Response to Comment 2
We agree that the physical activity measures used in our study represent heterogeneous operationalizations of the construct. However, the number of studies contributing to individual physical activity metrics was limited: n = 3 for objective steps per day, n = 1 for objective active time, n = 3 for subjective active time, n = 1 for subjective questionnaire score, and n = 12 for subjective MET-min/week. Moreover, these studies did not all contribute to the same TMS-derived outcomes, further reducing the number of studies available for any given combination of physical activity metric and neurophysiological outcome (n = 16 for CSE, n = 11 for SICI, n = 2 for cSP and iSP, n = 8 for ICF, and n = 3 for SICF). Accordingly, further subdivision by specific physical activity metric would result in estimates based on very few studies and would not provide robust evidence. We therefore retained the subgroup analyses focusing on the more meaningful distinction between self-reported (n = 16) and device-based measures (n = 4). These analyses did not provide evidence that the type of physical activity measurement altered the association with any of the TMS-derived outcomes. This point is acknowledged in the Limitations section.

> Page 13, lines 53-55: "*Second, habitual physical activity was assessed using different measures across studies. However, the small number of studies contributing to specific physical activity measures precluded more detailed analyses*."

Response to Comment 3

The search strategy was deliberately broad and included several terms related to physical activity, including physical activity, physical education, training, physical fitness, exercise, cycling, and treadmill. Therefore, studies involving strength or resistance training, aerobic training, or chronic training adaptations would have been identified by our search strategy and considered for inclusion if they met the eligibility criteria and examined the association between habitual physical activity and TMS-derived measures of motor neurophysiology, which was the focus of the present review. Studies examining only the acute effects of a single exercise bout were outside the scope of the present review, as these effects have been addressed in previous systematic reviews (e.g., Youssef et al., 2024). Studies focusing exclusively on laboratory-based measures of physical fitness were also excluded according to our eligibility criteria. We have clarified these distinctions in the description of the study eligibility criteria.

> Page 2, lines 11-18 (Introduction): "*Physical activity, which includes exercise, sport, active transport, household chores, and occupational activities (Bull et al. 2020), may influence M1 plasticity. Evidence from systematic reviews and meta-analyses indicates that acute exercise can modulate several neurophysiological measures of motor system function, including CSE as well as intracortical inhibitory and facilitatory circuits (Mellow et al. 2020; Youssef et al. 2024). These acute effects, typically observed within minutes following exercise, may be mediated by transient neurochemical changes, including increases in circulating catecholamines and neurotrophic factors (Rasmussen et al. 2009; Skriver et al. 2014). Whether repeated exposure to physical activity is associated with longer-term adaptations in resting M1 neurophysiology remains unclear.*"
>
> Page 3, lines 13-15 (Eligibility Criteria): "*Studies were included if they […] included at least one TMS-derived neurophysiological measure of motor system function and at least one measure of habitual physical activity, […].*"
>
> Page 3, lines 20-24 (Eligibility Criteria): "*Studies were excluded if they were published as a review article, book chapter, study protocol, or conference abstract, or assessed laboratory-based measures of physical fitness (e.g., maximal muscle force). Studies assessing only acute changes in TMS-derived measures following an exercise bout, without reporting a measure of habitual physical activity, were excluded.*"

Response to Comment 4

This comment has been addressed in our responses above, and the heterogeneity in the measurement of habitual physical activity is now explicitly acknowledged as a limitation of the present review.

> Page 13, lines 31-34 (Discussion): "*Taken together, these results do not provide evidence of a robust association between habitual physical activity and resting TMS-derived neurophysiological measures. However, the available evidence is*

insufficient to exclude measure-specific associations, particularly for some inhibitory measures for which relatively few studies were available."

Page 13, lines 53-55 (Discussion): "*Second, habitual physical activity was assessed using different measures across studies. However, the small number of studies contributing to specific physical activity measures precluded more detailed analyses.*"

Response to Comment 5

We agree that the interpretation of the inhibitory measures required greater caution. We have substantially revised and tempered the Discussion to emphasize that the available evidence does not show a consistent association across the different inhibitory measures, while acknowledging that the limited number of studies and physiological heterogeneity in some of these measures preclude ruling out measure-specific associations. In particular, we have substantially softened the interpretation of the iSP finding, which was based on only two effect sizes, and now describe it as hypothesis-generating rather than as evidence of a specific association between habitual physical activity and interhemispheric inhibition.

Page 13, lines 21-34 (Discussion): "*For inhibitory measures, the pooled association was non-significant and practically negligible, with moderate heterogeneity and a wide prediction interval, indicating substantial uncertainty about the association in future studies. Subgroup analyses revealed differences across inhibitory measures, with a positive association observed for iSP only. However, the iSP estimate was based on only two effect sizes and should therefore be considered hypothesis-generating rather than evidence of a specific association between habitual physical activity and interhemispheric inhibition. Similarly, the available evidence did not support a statistically significant association with facilitatory measures, although equivalence to zero could not be established and small associations cannot be ruled out.*
Taken together, these results do not provide evidence of a robust association between habitual physical activity and resting TMS-derived neurophysiological measures. However, the available evidence is insufficient to exclude measure-specific associations, particularly for some inhibitory measures for which relatively few studies were available."

Response to Comment 6

We agree that the Downs and Black checklist does not capture all methodological factors specific to TMS. We retained it as a broad assessment of methodological quality and, as noted in the Limitations, acknowledge that variability in TMS methodologies across studies may have contributed to heterogeneity in the reported neurophysiological measures.

Page 13, line 55 – Page 14 line 5 (Limitations): "*Second, variability in TMS methodologies across studies (e.g., differences in stimulation parameters,*

interstimulus intervals, and participant characteristics) may have contributed to heterogeneity in the reported neurophysiological measures."

Response to Comment 7

We agree that the broad search strategy included terms that could identify studies focused on acute exercise. These terms were intentionally included to maximize the sensitivity of the search and avoid missing studies that may have also reported an association between habitual physical activity and TMS-derived measures. The presence of these terms in the search strategy therefore does not imply that acute exercise was an eligible exposure. We have clarified in the eligibility criteria that studies were eligible only when they assessed habitual physical activity and formally tested its association with TMS-derived measures; studies examining only the acute effects of an exercise bout were not eligible.

> Page 3, lines 13-16 (Eligibility Criteria): "*Studies were included if they […] included at least one TMS-derived neurophysiological measure of motor system function and at least one measure of habitual physical activity, and formally quantified the association between these variables by reporting a Pearson's r coefficient or sufficient information to calculate it, […]*"
>
> Page 3, lines 23-24 (Eligibility Criteria): "*Studies assessing only acute changes in TMS-derived measures following an exercise bout, without reporting a measure of habitual physical activity, were excluded.*"

Response to Comment 8

Authors: Our eligibility criterion was based on whether studies formally tested the association between habitual physical activity and a TMS-derived measure, rather than on whether habitual physical activity was the primary or secondary objective of the original study. Thus, studies primarily designed to investigate acute exercise or other aspects of motor neurophysiology were eligible when they also reported an eligible association between habitual physical activity and a TMS-derived measure. We have clarified this distinction in the eligibility criteria and now clearly outline that none of the studies tested the association between habitual physical activity associations.

> Page 1, lines 24-26 (Abstract): "*We recovered previously unreported association estimates using statistical information from the articles, open datasets, and data requests to authors*."
>
> Page 3, line 17 (Eligibility Criteria): "*Studies were included […] irrespective of whether habitual physical activity was a primary exposure of interest*."
>
> Page 13, lines 4-10 (Discussion): "*An important strength of this study is that among the articles that collected measures of both habitual physical activity and TMS-derived neurophysiology, none reported estimates of the associations between these variables. By recovering these estimates from reported statistical information, open*

datasets, and data requests to authors, we performed secondary analyses of previously unanalyzed data and synthesized the resulting associations across studies, generating new empirical evidence on a question that had not previously been examined."

Response to Comment 9

We agree that the rationale for examining resting TMS-derived measures could be made more explicit. We have revised the Introduction to distinguish the transient effects of acute exercise from potential longer-term adaptations associated with habitual physical activity and to clarify that evidence for effects on resting neurophysiology remains limited. We now emphasize that repeated exposure to physical activity could, in principle, produce longer-lasting adaptations in neural systems involved in motor control and plasticity, while acknowledging that whether such adaptations are reflected in resting TMS-derived measures remains unclear.

> Page 2, lines 19-31 (Introduction): "Although the association between habitual physical activity, reflecting typical levels of physical activity accumulated across daily life (Bull et al. 2020), and resting M1 neurophysiology has not been systematically reviewed, several lines of evidence provide a rationale for investigating this relationship. Repeated exposure to physical activity may induce longer-lasting adaptations in neural systems involved in motor control and plasticity, potentially resulting in differences in resting neurophysiological measures (El-Sayes et al. 2019). For example, habitual physical activity has been associated with alterations in neural systems supporting plasticity, including increases in circulating neurotrophic factors such as brain-derived neurotrophic factor (BDNF) (Rasmussen et al. 2009). Moreover, neurophysiological responses to acute exercise appear to vary according to habitual physical activity levels (Cirillo et al. 2009; Lulic et al. 2017), raising the possibility that habitual physical activity may be associated not only with responsiveness to acute exercise but also with the baseline state of the motor system. However, whether habitual physical activity is associated with resting TMS-derived measures remains unclear."

Response to Comment 10

We agree that the term "motor cortical physiology" is broader than is appropriate for all of the neurophysiological measures included in the review. We have therefore revised the terminology throughout the manuscript to distinguish corticospinal excitability, assessed primarily using MEPs elicited by single-pulse TMS, from TMS-derived measures of intracortical and interhemispheric inhibitory and facilitatory processes. We now use "TMS-derived neurophysiological measures" as the overarching term where appropriate.

Response to Comment 11

We agree that the conclusion should be more restrained. We have revised the conclusion to avoid broad statements about "global" differences in motor cortical neurophysiology and to more

precisely reflect the TMS-derived measures examined in this review. The revised conclusion now states:

> Page 14, lines 8-19 (Conclusion): "Overall, current evidence does not support a robust association between habitual physical activity and resting TMS-derived measures of CSE, intracortical inhibition (SICI), or intracortical facilitation (ICF and SICF). LICI, cSP, and iSP were examined in only one or two studies, limiting conclusions about these measures. Although some associations could not be considered equivalent to zero, further research should be pursued only where there is a clear theoretical rationale and with sufficient methodological rigor, including objective assessment of habitual physical activity (e.g., accelerometry), detailed TMS protocols, preregistration, and open sharing of data and analysis scripts. Future studies could also determine whether associations emerge beyond resting TMS-derived measures, for example during motor tasks involving approach–avoidance behavior (Park and Boisgontier 2025), or whether habitual physical activity is associated with broader motor-network interactions, including those involving regions implicated in the perception of physical effort (Zénon et al. 2015)."

REVIEWER 2

SUMMARY

The authors conducted a systematic review and meta-analysis of the literature on (acute) physical activity and transcranial magnetic stimulation (TMS) measures of corticospinal motor function, including MEP amplitudes, EMG fluctuations ('silent periods') and a range of paired-pulse measures. As with all attempts to synthesize a research literature (a process similar to 'herding cats'), the necessary data was available only for a subset of relevant studies, and the necessary variables were similarly restricted. However, meta-analysis over the available data revealed no strong effects of physical activity on any measure, and some small effects on some measures. The strongest evidence here shows no effects of acute physical activity on corticospinal function.

EVALUATION

This is excellent work.

Authors: We thank Reviewer 2 for the positive evaluation of our work.

The systematic review and meta-analysis methods and results are excellent, very thorough, using the best-available tools and procedure. This is better than I would do myself, and I have only a few minor suggestions about phrasing in a few places. At multiple points I found myself making comments only to delete them later as the authors had already addressed them in the text.

Authors: We sincerely thank Reviewer 2 for this exceptionally positive and encouraging assessment of our work. Positive feedback of this kind is not always common in the peer-review process, and we particularly appreciate the recognition of the methodological rigor and thoroughness of the review. This encouragement is especially meaningful to the student who led the work and manuscript.

My largest concern is with the 'Limitations' section, which goes far beyond what is needed, and even contradicts some of the conclusions made in the preceding text. I would strongly suggest reducing this to the absolute minimum and presenting instead a bolder, more confident conclusion.

Authors: As recommended and detailed in our responses below, we have reduced the Limitations section.

> Page 13, line 50 – Page 14, line 5 (Limitations): "*Several limitations should be considered when interpreting the results. First, most studies assessing physical activity relied on self-reported questionnaires, which may be subject to recall bias and measurement error compared with device-based approaches such as accelerometry. Second, habitual physical activity was assessed using different*

measures across studies. However, the small number of studies contributing to specific physical activity measures precluded more detailed analyses. Third, variability in TMS methodologies across studies (e.g., differences in stimulation parameters, interstimulus intervals, and participant characteristics) may have contributed to heterogeneity in the reported neurophysiological measures. Fourth, for paired-pulse measures, restricted between-participant variability in the derived measures, if present, could attenuate correlations with habitual physical activity."

SPECIFIC MAJOR CONCERNS

None

Authors: Thank you.

SPECIFIC MINOR CONCERNS

Abstract

- I'm generally uncomfortable with statistical results in abstracts because i fear that is all people will read. However, since these are standardised effect sizes with confidence intervals and no p-values, this is good.

Authors: Statistical results have been removed from the abstract to allow space for other important information raised by the reviewers.

- "Subgroup analyses... no significant difference" - plural differenceS? Or "no significant effects"?

Authors: Done.

Introduction

- Line 36: "descending corticospinal signals" - is it worth mentioning other non-spinal targets of M1 outputs? - cortico-cortical, pontine, brainstem, etc? (e.g., a quick search led me to this report of the bewilderingly-well-connected motor areas in mice: https://doi.org/10.1038/srep20072)

Authors: We agree that M1 has outputs to multiple cortical and subcortical regions in addition to the corticospinal pathway. We have therefore revised the sentence to avoid implying that corticospinal projections represent the exclusive output of M1, while retaining the focus on corticospinal signals relevant to the TMS measures examined in this review.

Page 2, lines 3-6 (Introduction): "*The primary motor cortex (M1) plays a key role in the control of voluntary movement by integrating inputs from multiple cortical and*

subcortical regions and contributing to motor output through descending corticospinal signals that drive skeletal muscle activity (Strick et al. 2021)."

- Line 61 "cortical silent period (cSP)" - are we sure this is entirely cortical (and not mostly, eg, spinal motorneuronal) - relatively few studies find SP without an MEP I believe(?), though other SPs can be elicited (e.g. https://doi.org/10.1016/j.brs.2026.103105)? I appreciate it is called the cortical SP - perhaps a rebranding to 'contralateral' SP or 'corticospinal' SP is needed here?

Authors: We thank Reviewer 2 for raising this point. We agree that the cSP should not be interpreted as an exclusively cortical measure, as both cortical and spinal mechanisms contribute to the silent period. We now describe the cSP as reflecting predominantly cortical inhibition while acknowledging contributions from spinal mechanisms. After careful consideration, we also decided to use the term "contralateral silent period" as this terminology more accurately avoids implying an exclusively cortical origin.

> Page 2, lines 35-38 (Introduction): "*Inhibitory processes can also be assessed using the contralateral silent period (cSP) and the ipsilateral silent period (iSP), which predominantly reflect cortical inhibition and interhemispheric inhibition, respectively, although the cSP also includes contributions from spinal mechanisms (Hupfeld et al. 2020).*"

Methods

- Line 80 "To reduce potential bias" - I can see that pre-registering allows others to compare the planned work with the eventual work, but preregistration alone is not a bias-reduction method. Rephrase?

Authors: Agreed. We have removed "To reduce potential bias" and now simply state "*To enhance transparency, this systematic review was pre-registered […]*"

- Line 89 "by reporting a Pearson's r correlation coefficient" - perhaps say more about why this was essential? All other effect sizes can be converted to r, so the requirement for a correlation to be reported limits this study to 'individual differences' rather than - eg - discrete or group-level intervention effects. That was not clear to me until this point, and may need reflecting in the title, abstract etc. (After reading lines 128+, I am no longer clear about this.)

Authors: We agree that the rationale for requiring a correlation or equivalent statistic should be made clearer. Our objective was specifically to examine whether individual differences in habitual physical activity are associated with individual differences in TMS-derived neurophysiological measures. Accordingly, studies were required to report a statistic quantifying this association, such as Pearson's r or an equivalent effect size that could be converted to r. This criterion was not intended to exclude studies on the basis of the statistical method used; rather, it ensured that the

synthesized effects directly addressed our research question. Group-level intervention effects, including changes following an exercise-training intervention, address a different question and were therefore outside the scope of this review. After careful consideration, we decided not to revise the abstract, as its explicit focus on an association already clearly signals that the study examines an individual differences.

> Page 3, lines 13-16 (Eligibility Criteria): "*Studies were included if they […] included at least one TMS-derived neurophysiological measure of motor system function and at least one measure of habitual physical activity, and formally quantified the association between these variables by reporting a Pearson's r coefficient or sufficient information to calculate it, […]*"

- Line 112 "...outcome measures, TMS measures..." - as an experienced reviewer of the TMS literature, I know that this list could easily be ~200 variables. Could the authors please provide numbers on these metadata variables extracted? Was this all decided at the start of the study, or did multiple revisions and re-assessments get made as the studies accumulated and new variables were discovered?

Authors: We thank Reviewer 2 for this important point. Five TMS-derived outcome measures were prespecified in the preregistration (CSE, SICI, LICI, ICF, and SICF). These were selected a priori based on the established TMS literature and previous reviews (Alibazi et al., 2021; Youssef et al., 2024) to represent corticospinal excitability, intracortical inhibition, and intracortical facilitation. During full-text screening and data extraction, we identified the silent period as another relevant TMS-derived measure of inhibition that fell within the conceptual scope of the review. We therefore added the contralateral silent period (cSP) and ipsilateral silent period (iSP) as additional inhibitory outcomes. This decision was made during study selection and data extraction, before quantitative analyses. For the remaining study characteristics and methodological variables, we did not restrict extraction to a prespecified list. Rather, we extracted all relevant physical activity and TMS-derived variables reported by eligible studies. Consequently, the number and nature of these variables varied across studies according to what was reported. We have clarified this in the Methods and now explicitly distinguish the preregistered outcomes from variables identified during data extraction.

> Page 3, lines 43-48 (Data Extraction): "*Five TMS-derived outcome measures were prespecified in the PROSPERO preregistration (CSE, SICI, LICI, ICF, and SICF). During study selection and data extraction, cSP and iSP were added as relevant inhibitory outcomes that fell within the conceptual scope of the review. Habitual physical activity and other TMS-related variables were not restricted to a prespecified list; all relevant variables reported by eligible studies were extracted*."

References:

- Alibazi RJ, Pearce AJ, Rostami M, Frazer AK, Brownstein C, Kidgell DJ. Determining the intracortical responses after a single session of aerobic exercise in young healthy individuals: a systematic review and best evidence synthesis. J Strength Cond Res. 2021;35(2):562-575.

- Youssef L, Harroum N, Francisco BA, et al. Neurophysiological effects of acute aerobic exercise in young adults: a systematic review and meta-analysis. Neurosci Biobehav Rev. 2024;164:105811.

- Line 137+ - there are primary, sub-group, and meta-regression analyses - approximately how many analyses were performed - e.g., there 6 TMS measures are reported but others must have been found? Is it worth saying which analyses were pre-registered?

Authors: Our preregistration did not prespecify the individual meta-analytic models because the feasibility and structure of the quantitative synthesis depended on the data available from the eligible studies; it stated that a meta-analysis would be conducted if sufficient data could be gathered. As mentioned above, the main TMS-derived outcomes included in the review were prespecified and for each of them, we tested the associations with all habitual physical activity variables reported by the included studies. Further, the analytical approach was highly similar to that used in three previous meta-analyses from our group. Thus, the overall analytical framework was established independently of the present results and was not developed post hoc to explain or emphasize particular results. We have clarified this analytical framework and the distinction between prespecified outcomes and data-dependent analyses in the Methods.

> Page 3, lines 43-48 (Methods): "*Five TMS-derived outcome measures were prespecified in the PROSPERO preregistration (CSE, SICI, LICI, ICF, and SICF). During study selection and data extraction, cSP and iSP were added as relevant inhibitory outcomes that fell within the conceptual scope of the review. Habitual physical activity and other TMS-related variables were not restricted to a prespecified list; all relevant variables reported by eligible studies were extracted*."

> Page 4, lines 8-10 (Methods): "*The meta-analytic approach followed the framework used in our previous systematic reviews and meta-analyses (Goubran et al. 2025; Farajzadeh et al. 2026a; Jabouille et al. 2026), with the specific outcomes and associations determined by the data available from eligible studies*."

Should we correct across the meta-analyses for multiple comparisons?

Authors: We did not apply a global multiple-comparison correction across the primary meta-analyses because these addressed distinct neurophysiological outcomes (CSE, inhibitory processes, facilitatory processes). The inhibitory and facilitatory measures are also not interchangeable measures as they probe partially distinct neurophysiological processes. We therefore did not apply a correction across subgroup analyses either, but interpret their findings cautiously, particularly where few studies contributed to a subgroup. Applying a conventional multiple-comparison correction would not have substantially altered the interpretation of most analyses, as the pooled estimates for CSE, inhibitory processes, and facilitatory processes, as well as the measure-specific estimates for SICI, LICI, ICF, SICF, and cSP, were already not statistically significant.

Pages 4, lines 51-55 (Methods): "*Subgroup analyses were conducted to examine whether pooled correlations differed across types of inhibitory (SICI, LICI, cSP, and iSP) and facilitatory measures (ICF and SICF), as well as by physical activity assessment method (self-reported vs. device-based). The inhibitory measures (SICI, LICI, cSP, and iSP) probe partially distinct neurophysiological processes and should not be interpreted as interchangeable measures of a single inhibitory mechanism. Thus, the overall pooled estimate for inhibitory measures represents a statistical summary across these measures, whereas the subgroup estimates provide measure-specific associations. Similarly, ICF and SICF reflect distinct forms of intracortical facilitation.*"

Results

-Table 1:

a) %female (I assume the remainder are male - add note to the table); Mean+SD - is this age in years? Muscle - give definitions in table notes.

Authors: Done.

b) Suggest using separate columns for all numbers - this really helps when copying others' tables from the internet/pdf; and it is also much easier to read when similar numbers are aligned; so, separate columns for (% female), M and SD, Activity/M/SD please.

Authors: Done.

c) Should r values be accompanied with degrees-of-freedom and/or SE? I know these should be derivable directly from N and r, but it might be worth adding, as effect-size SEs are hard to track down (https://matthewbjane.quarto.pub/guide-to-effect-sizes-and-confidence-intervals/).

Authors: We agree that reporting effect-size precision facilitates interpretation and reproducibility. We have therefore added a column to Table 1 reporting the standard error (SE) of each correlation after Fisher's r-to-z transformation.

Page 8, line 42 (Table note): "SEz = standard error of the Fisher's z-transformed correlation coefficient calculated as 1/(N−3)"

d) Some quality scores are underlined - what does this mean?

Authors: The underlining of some quality scores was a formatting error. We have removed it.

e) "CSE (110; postero-anterior)" - 110% (RMT?) and PA can be reported for all studies in this table. Please reconsider which TMS variables need to be in Table 1 and be consistent for all studies; e.g. SICI will be tested with different conditioning and test pulse intensities; likewise all the other protocols.

Authors: To be consistent across studies, we have removed protocol-specific details from CSE entries. We retained the ISI column for the paired-pulse protocols.

- Line 294 "(total I2 of 40.99%)" - this doesn't match up with I^2 in the Figure 3 (36.4%), please check/explain in text?

Authors: This was a typo in the text, thank you.

Discussion
- Line 385 "g = 0.27" - please explain this effect size - Hedges?

Authors: The reviewer is correct, it is Hedges g. We have now clarified it in the text.

> Page 13, line 39 (Discussion): "*Specifically, acute aerobic exercise produced a small increase in intracortical inhibition, reflected by a decrease in SICI (Hedges' g = 0.27),* […]"

Limitations
I appreciate the authors' inclusion of a limitations section, but I would encourage them to think more about some of them, and to be more confident / bold in their conclusions. Some details:

Authors: We tend to be comprehensive in describing the limitations of our studies, but we agree that we were overly cautious in this case. We have therefore revised the Limitations section to provide a more balanced interpretation of the findings.

> Page 13, line 50 – Page 14, line 5 (Limitations): "*Several limitations should be considered when interpreting the results. First, most studies assessing physical activity relied on self-reported questionnaires, which may be subject to recall bias and measurement error compared with device-based approaches such as accelerometry. Second, habitual physical activity was assessed using different measures across studies. However, the small number of studies contributing to specific physical activity measures precluded more detailed analyses. Third, variability in TMS methodologies across studies (e.g., differences in stimulation parameters, interstimulus intervals, and participant characteristics) may have contributed to heterogeneity in the reported neurophysiological measures. Fourth, for paired-pulse measures, restricted between-participant variability in the derived measures, if present, could attenuate correlations with habitual physical activity.*"

-"First, the analyses were based on correlational data and therefore do not permit conclusions regarding causal relationships" - I have never really understood this 'limitation', which often appears when r-values are used as effect sizes: yes, the data (all data!) are 'correlational', but r can be (and was) derived from and related to all other statistical effect sizes (t, F, or p) - so it's not clear why 'correlation is not equal to causation' here. This claim really requires an analysis of the causal theory underlying all the individual studies - I imagine that some studies will be more causal than others. For example, if I force people (on a large hamster wheel in a climate-controlled exercise lab) to do different levels of exercise, and find that activity level correlates ~perfectly with corticospinal excitability, I will be quite content to say that a causal relationship has been established. (It cannot be the opposite, that CSE causes activity level, for example.)

Authors: Thank you for this insightful perspective. We had not previously considered this point from this angle, and this comment led us to reconsider whether the use of correlation coefficients as effect sizes should be presented as a limitation. We agree that it should not, because it is not, in itself, evidence that the underlying studies are non-causal. We have therefore removed this limitation from the manuscript.

- Limitations 1 and 4 can be combined (or #1 can be removed).

Authors: As recommended above, limitation #1 was removed.

- Line 419 "differences in stimulation parameters, interstimulus intervals, and participant characteristics) may have contributed to heterogeneity in the reported neurophysiological measures." - some (or all?) of these can be assessed post-hoc with the current data. Most studies report stimulation parameters, ISIs and age/sex/handedness of participants. Please do the analysis and check for these; report in a post-hoc section of the results ("Prompted by a reviewer, we also analysed...".)

Authors: We thank Reviewer 2 for this suggestion. We examined the available data for participant characteristics, ISIs, and TMS stimulation parameters to determine whether these variables could be meaningfully evaluated as potential moderators. Age and sex were already examined in our meta-regression analyses, with no statistically significant moderator effects, whereas the remaining variables did not provide sufficiently large or balanced groups for reliable post-hoc moderator analyses.

Although ISIs were reported, their distributions did not provide sufficiently large or balanced groups for reliable moderator analyses. SICI was assessed at 2 ms in all but two studies, with ISIs of 2.5 and 3 ms each represented by only one study. ICF was assessed at 10 ms in four studies and at 12 ms in two studies. SICF was assessed at 1.5 ms in two studies, whereas estimates at 1.2 and 2.5 ms came from the same single study. LICI at 100 ms was available from only one study. Thus, subgroup analyses based on ISI would have resulted in very small and uneven groups, and meta-regression would not have provided a reliable estimate of an ISI effect.

Handedness was reported in 16 of the 20 included studies and was not reported in four. Fourteen studies included exclusively right-handed participants. One study included 19 right-handed and one left-handed participant, and another included 13 right-handed and three participants of mixed handedness. Thus, no study included a predominantly left-handed or mixed-handed sample, precluding a meaningful analysis of handedness as a moderator. However, we have added a description of handedness in the text and a column to Table 1 reporting the percentage of right-handed participants in each study.

As pointed out by this reviewer, TMS stimulation parameters encompass numerous technical and methodological factors, including stimulation intensity, conditioning- and test-stimulus intensities, coil type, pulse waveform, coil orientation and induced-current direction, number of stimuli, muscle activation state, and target muscle. We reviewed the available stimulation characteristics to determine whether any parameters provided sufficiently populated and distinct categories for analysis. However, parameters were represented by only one or a few independent studies, or occurred as multiple experimental conditions within the same study. For example, most studies assessed the FDI (n = 17), whereas APB (n = 3), TA (n = 1), and VL (n = 2) were each represented by only a small number of studies. Consequently, the available evidence was too limited to support reliable moderator of individual TMS stimulation parameters.

Page 6 line 51 – Page 8, line 13 (Descriptive Results): "*Eighteen studies assessed TMS measures in upper-extremity muscles (Cirillo et al. 2009; McGregor et al. 2011; McGregor et al. 2013; Lulic et al. 2017; Houde et al. 2018; El-Sayes et al. 2019; Rozand et al. 2019; Neva et al. 2021; Smith et al. 2021; Hu et al. 2021; Hand et al. 2022b; Wang et al. 2023; Curtin et al. 2023; Ross et al. 2024; Digonet et al. 2025; Harroum et al. 2025; Gamage et al. 2026; Farajzadeh 2026) (Table 1). Two studies examined both upper- and lower-extremity muscles (Hand et al. 2022a; Hougland et al. 2025), and one examined only lower-extremity muscles (Hassanlouei et al. 2017). Handedness was not reported in four studies (Hassanlouei et al. 2017; Smith et al. 2021; Hand et al. 2022a; Hand et al. 2022b). Fifteen studies included exclusively right-handed participants (Cirillo et al. 2009; McGregor et al. 2011; McGregor et al. 2013; Lulic et al. 2017; Houde et al. 2018; El-Sayes et al. 2019; Rozand et al. 2019; Hu et al. 2021; Neva et al. 2021; Wang et al. 2023; Hougland et al. 2025; Digonet et al. 2025; Harroum et al. 2025; Gamage et al. 2026; Farajzadeh 2026). One study included 19 right-handed and one left-handed participants (Curtin et al. 2023), another study included 13 right-handed and three mixed-handed participants (Ross et al. 2024).*

CSE was assessed in sixteen studies and reported as raw MEP amplitude (Cirillo et al. 2009; Houde et al. 2018; Neval et al. 2021; Smith et al. 2021; Hu et al. 2021; Ross et al. 2024; Harroum et al. 2025; Farajzadeh 2026; Hand et al. 2022b), input–output curve slope (Wang et al. 2023; Hougland et al. 2025), and area under the recruitment curve (Lulic et al. 2017). In other studies, M-wave-normalized CSE was reported as MEP amplitude (Digonet et al. 2025), area under the curve (El-Sayes et al. 2019), or MEPmax (Hassanlouei et al. 2017; Rozand et al. 2019), all expressed relative to Mmax. SICI was assessed in twelve studies, using an interstimulus interval (ISI) of 2 ms (Cirillo et al. 2009; Lulic et al. 2017; Neva et al. 2021; Hand et al. 2022a; El-Sayes et al. 2019; Hand et al. 2022b; Wang et al. 2023; Digonet et al. 2025; Harroum et al. 2025; Farajzadeh 2026), 2.5 ms (Hu et al. 2021), or 3 ms (Digonet et al. 2025;

Neva et al. 2021). LICI was assessed in one study using an ISI of 100 ms (Farajzadeh 2026). cSP was assessed in two studies (Cirillo et al. 2009; Houde et al. 2018) and iSP in two studies (McGregor et al. 2011; McGregor et al. 2013). ICF was assessed in seven studies, with ISI of 10 ms in four studies (Lulic et al. 2017; Hand et al. 2022a; Hand et al. 2022b; Wang et al. 2023) and 12 ms in three studies (Hu et al. 2021; Curtin et al. 2023; Farajzadeh 2026). SICF was assessed in three studies using ISIs of 1.2 ms (Lulic et al. 2017), 1.5 ms (Hand et al. 2022a; Hand et al. 2022b), and 2.5 ms (Lulic et al. 2017). Some studies assessed TMS measures using both antero-posterior and postero-anterior current directions (Neva et al. 2021; Harroum et al., 2025)."

- Line 421 "Future research using larger samples... will be important to better clarify the relationship between habitual physical activity and motor cortical physiology." - I disagree. If the effects are truly (or practically) zero, then no sample size will ever be sufficient to establish or even clarify such a relationship. If the studies included were good (or if study quality was irrelevant), zero effects implies that no further research needs to be done: https://deevybee.blogspot.com/2024/03/just-make-it-stop-when-will-we-say-that.html

Authors: Thank you for this thoughtful comment and for pointing us to Dorothy Bishop's blog post. After considering this point further, we agree that recommending larger samples alone may imply that insufficient statistical power is the primary limitation, whereas our findings suggest that the associations examined may be very small or practically negligible. We have therefore revised the conclusion to no longer recommend larger samples as a general priority. Instead, we now distinguish between the measures for which the available evidence was more substantial (CSE, SICI, ICF, and SICF) and those for which evidence remains limited (cSP, iSP, and LICI), and we identify more targeted directions for future research, including examining responses during motor tasks and broader motor-network interactions.

Page 14, lines 8-19 (Conclusion): "Overall, current evidence does not support a robust association between habitual physical activity and resting TMS-derived measures of CSE, intracortical inhibition (SICI), or intracortical facilitation (ICF and SICF). LICI, cSP, and iSP were examined in only one or two studies, limiting conclusions about these measures. […] Future studies could also determine whether associations emerge beyond resting TMS-derived measures, for example during motor tasks involving approach–avoidance behavior (Park and Boisgontier 2025), or whether habitual physical activity is associated with broader motor-network interactions, including those involving regions implicated in the perception of physical effort (Zénon et al. 2015)."

- For the effects that did not have a significant equivalence test, it may make some sense to cautiously suggest some limited further study, but given the lack of definitive effects, future research should only be done with significantly improved: theory, methods, and analysis, and fully-open registration and data sharing.

Authors: We agree that the absence of significant equivalence tests for some associations warrants cautious interpretation rather than a general recommendation for further research. We have therefore revised the conclusion to emphasize that future research should be pursued only where there is a clear theoretical rationale and with substantially improved methodological rigor.

> Page 14, lines 12-15 (Conclusion): "Although some associations could not be considered equivalent to zero, further research should be pursued only where there is a clear theoretical rationale and with sufficient methodological rigor, including objective assessment of habitual physical activity (e.g., accelerometry), detailed TMS protocols, preregistration, and open sharing of data and analysis scripts."

Signed: Nick Holmes (I sign all my reviews)

REVIEWER 3

Response to Comment 1

We agree that the method and time window used to assess habitual physical activity, as well as the methodological characteristics of the TMS measurements, are important when interpreting between-participant associations. The physical activity measures used across the included studies were not uniformly limited to the preceding week; assessment periods and methods varied across questionnaires and device-based approaches. We have clarified these characteristics in the manuscript.

We agree that the physical activity measures used in our study represent heterogeneous operationalizations of the construct. However, the number of studies contributing to individual physical activity metrics was limited: n = 3 for objective steps per day, n = 1 for objective active time, n = 3 for subjective active time, n = 1 for subjective questionnaire score, and n = 12 for subjective MET-min/week. Moreover, these studies did not all contribute to the same TMS-derived outcomes, further reducing the number of studies available for any given combination of physical activity metric and neurophysiological outcome (n = 16 for CSE, n = 11 for SICI, n = 2 for cSP and iSP, n = 8 for ICF, and n = 3 for SICF). Accordingly, further subdivision by specific physical activity metric would result in estimates based on very few studies and would not provide robust evidence. We therefore retained the subgroup analyses focusing on the more meaningful distinction between self-reported (n = 16) and device-based measures (n = 4). This point is acknowledged in the Limitations section.

> Page 6, lines 37-49 (Descriptive Results): "*Four studies assessed physical activity using triaxial accelerometers (Hassanlouei et al. 2017; Rozand et al. 2019; Smith et al. 2021; Farajzadeh 2026), which were worn at the waist (ActiGraph GT3X) (Hassanlouei et al. 2017; Rozand et al. 2019; Farajzadeh 2026) or the wrist (ActivInsights GENEActiv) (Smith et al. 2021) (Table 1). Seventeen studies assessed physical activity using self-reported measures. Thirteen of these used the International Physical Activity Questionnaire (Cirillo et al. 2009; Lulic et al. 2017; El-Sayes et al. 2019; Hu et al. 2021; Neva et al. 2021; Hand et al. 2022a; Hand et al. 2022b; Curtin et al. 2023; Ross et al. 2024; Digonet et al. 2025; Harroum et al. 2025; Hougland et al. 2025; Gamage et al. 2026). Other self-report instruments were the Leisure Time Exercise Questionnaire (McGregor et al. 2011), the Physical Activity Questionnaire (Wang et al. 2023), and ad hoc questionnaires (McGregor et al. 2013; Houde et al. 2018).*
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> *Regarding the assessment period, accelerometer-based physical activity was assessed over 7 days in one study (Smith et al. 2021), 5 days in one study (Farajzadeh 2026), and 4 days in two studies (Hassanlouei et al. 2017; Rozand et al. 2019). Self-reported physical activity generally referred to the previous 7 days (Cirillo et al. 2009; McGregor et al. 2011; McGregor et al. 2013; Lulic et al. 2017; El-Sayes et al. 2019; Hu et al. 2021; Neva et al. 2021; Hand et al. 2022a; Hand et al. 2022b; Curtin et al. 2023; Ross et al. 2024; Digonet et al. 2025; Harroum et al. 2025; Hougland et al. 2025; Gamage et al. 2026), with*

one study assessing activity over the previous 5 years (Wang et al. 2023) and one not reporting the assessment period (Houde et al. 2018).
Physical activity outcomes included MET-min/week (Cirillo et al. 2009; Lulic et al. 2017; El-Sayes et al. 2019; Hu et al. 2021; Neva et al. 2021; Hand et al. 2022a; Hand et al. 2022b; Curtin et al. 2023; Ross et al. 2024; Digonet et al. 2025; Harroum et al. 2025; Hougland et al. 2025; Gamage et al. 2026), steps per day (Hassanlouei et al. 2017; Rozand et al. 2019; Farajzadeh 2026), active time per week (McGregor et al. 2013; Houde et al. 2018; Smith et al. 2021; Wang et al. 2023), and questionnaire scores (McGregor et al. 2011)."

Page 13, line 55 – Page 14 line 5 (Limitations): "*Second, variability in TMS methodologies across studies (e.g., differences in stimulation parameters, interstimulus intervals, and participant characteristics) may have contributed to heterogeneity in the reported neurophysiological measures.*"

Response to Comment 2
Authors: We agree that adding greater detail on how the TMS-derived outcomes were obtained is important. We have therefore expanded the Methods to describe handedness, ISI for paired-pulse measures, and specific approach to measure CSE. As pointed out by the reviewer, TMS stimulation protocols encompass numerous technical and methodological factors. We reviewed the available stimulation characteristics to determine whether any provided sufficiently populated and distinct categories for analysis. In general, however, these parameters were represented by only one or a few independent studies, or occurred as multiple experimental conditions within the same study. Consequently, the available evidence was too limited to support reliable moderator analyses of individual TMS methodological characteristics.

Response to Comment 3
Authors: We agree that range restriction in TMS-derived measures can attenuate between-participant correlations. However, expressing conditioning stimulus intensity relative to individual motor thresholds does not necessarily result in restricted variability, and this concern does not directly apply to CSE. Restricted variability in paired-pulse measures, if present, could attenuate associations with habitual physical activity. Because individual-level distributions were not consistently available, we could not quantify the extent or impact of potential range restriction. We therefore consider this a potential source of attenuation rather than evidence of false-negative findings. This limitation is now acknowledged in the Discussion.

Page 14, lines 4-6 (Limitations): "*Fourth, for paired-pulse measures, restricted between-participant variability in the derived measures, if present, could attenuate correlations with habitual physical activity.*"

Response to Comment 4
Authors: We agree that methodological factors such as electrode placement, hotspot localization, background muscle activity, posture, and coil positioning can contribute to between-participant

variability in TMS-derived measures and may therefore affect the precision of between-participant association estimates. However, we do not consider these sources of variability to preclude quantitative synthesis. Rather, they represent potential sources of measurement error and between-study heterogeneity that should be considered when interpreting the magnitude and precision of the pooled associations. This limitation is acknowledged in the Discussion.

> Page 13, line 55 - Page 14, line 4 (Limitations): "*Third, variability in TMS methodologies across studies (e.g., differences in stimulation parameters, interstimulus intervals, and participant characteristics) may have contributed to heterogeneity in the reported neurophysiological measures*."

Response to Comment 5

Authors: We agree that adding greater detail on how the TMS-derived outcomes were obtained is important. We have therefore expanded the Methods to describe handedness, ISI for paired-pulse measures, and specific approach to measure CSE. As pointed out by the reviewer, TMS stimulation protocols encompass numerous technical and methodological factors. We reviewed the available stimulation characteristics to determine whether any provided sufficiently populated and distinct categories for analysis. In general, however, these parameters were represented by only one or a few independent studies, or occurred as multiple experimental conditions within the same study. Consequently, the available evidence was too limited to support reliable moderator analyses of individual TMS methodological characteristics.

> Page 6 line 51 – Page 8, line 13 (Descriptive Results): "*Eighteen studies assessed TMS measures in upper-extremity muscles (Cirillo et al. 2009; McGregor et al. 2011; McGregor et al. 2013; Lulic et al. 2017; Houde et al. 2018; El-Sayes et al. 2019; Rozand et al. 2019; Neva et al. 2021; Smith et al. 2021; Hu et al. 2021; Hand et al. 2022b; Wang et al. 2023; Curtin et al. 2023; Ross et al. 2024; Digonet et al. 2025; Harroum et al. 2025; Gamage et al. 2026; Farajzadeh 2026) (Table 1). Two studies examined both upper- and lower-extremity muscles (Hand et al. 2022a; Hougland et al. 2025), and one examined only lower-extremity muscles (Hassanlouei et al. 2017). Handedness was not reported in four studies (Hassanlouei et al. 2017; Smith et al. 2021; Hand et al. 2022a; Hand et al. 2022b). Fifteen studies included exclusively right-handed participants (Cirillo et al. 2009; McGregor et al. 2011; McGregor et al. 2013; Lulic et al. 2017; Houde et al. 2018; El-Sayes et al. 2019; Rozand et al. 2019; Hu et al. 2021; Neva et al. 2021; Wang et al. 2023; Hougland et al. 2025; Digonet et al. 2025; Harroum et al. 2025; Gamage et al. 2026; Farajzadeh 2026). One study included 19 right-handed and one left-handed participants (Curtin et al. 2023), another study included 13 right-handed and three mixed-handed participants (Ross et al. 2024).*
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> *CSE was assessed in sixteen studies and reported as raw MEP amplitude (Cirillo et al. 2009; Houde et al. 2018; Neval et al. 2021; Smith et al. 2021; Hu et al. 2021; Ross et al. 2024; Harroum et al. 2025; Farajzadeh 2026; Hand et al. 2022b), input–output curve slope (Wang et al. 2023; Hougland et al. 2025), and area under the recruitment curve (Lulic et al. 2017). In other studies, M-wave-normalized CSE was reported as MEP amplitude (Digonet et al. 2025), area under the curve (El-Sayes et al. 2019), or*

MEPmax (Hassanlouei et al. 2017; Rozand et al. 2019), all expressed relative to Mmax. SICI was assessed in twelve studies, using an interstimulus interval (ISI) of 2 ms (Cirillo et al. 2009; Lulic et al. 2017; Neva et al. 2021; Hand et al. 2022a; El-Sayes et al. 2019; Hand et al. 2022b; Wang et al. 2023; Digonet et al. 2025; Harroum et al. 2025; Farajzadeh 2026), 2.5 ms (Hu et al. 2021), or 3 ms (Digonet et al. 2025; Neva et al. 2021). LICI was assessed in one study using an ISI of 100 ms (Farajzadeh 2026). cSP was assessed in two studies (Cirillo et al. 2009; Houde et al. 2018) and iSP in two studies (McGregor et al. 2011; McGregor et al. 2013). ICF was assessed in seven studies, with ISI of 10 ms in four studies (Lulic et al. 2017; Hand et al. 2022a; Hand et al. 2022b; Wang et al. 2023) and 12 ms in three studies (Hu et al. 2021; Curtin et al. 2023; Farajzadeh 2026). SICF was assessed in three studies using ISIs of 1.2 ms (Lulic et al. 2017), 1.5 ms (Hand et al. 2022a; Hand et al. 2022b), and 2.5 ms (Lulic et al. 2017). Some studies assessed TMS measures using both antero-posterior and postero-anterior current directions (Neva et al. 2021; Harroum et al., 2025)."

Response to Comment 6

We agree that range restriction in TMS-derived measures can attenuate between-participant correlations. However, expressing conditioning stimulus intensity relative to individual motor thresholds does not necessarily result in restricted variability, and this concern does not directly apply to CSE. Restricted variability in paired-pulse measures, if present, could attenuate associations with habitual physical activity. Because individual-level distributions were not consistently available, we could not quantify the extent or impact of potential range restriction. We therefore consider this a potential source of attenuation rather than evidence of false-negative findings. This limitation is now acknowledged in the Discussion.

Page 14, lines 5-6 (Limitations): "*Fourth, for paired-pulse measures, restricted between-participant variability in the derived measures, if present, could attenuate correlations with habitual physical activity.*"

Response to Comment 7

We agree that the heterogeneity in TMS methodologies and physical activity measures should be considered when interpreting the findings, and we have explicitly acknowledged these limitations. However, we think that such heterogeneity does not preclude meaningful synthesis of the results for at least two reasons. First, documenting this heterogeneity is itself informative as it informs researcher that greater standardization or more detailed reporting is needed in future research. Second, the individual study estimates show a high degree of consistency with the pooled findings despite this methodological heterogeneity. Specifically, 19 of 20 CSE estimates, 12 of 13 SICI estimates, 2 of 2 cSP estimates, 1 of 1 LICI estimate, 9 of 9 ICF estimates, and 5 of 5 SICF estimates were statistically non-significant. Overall, 48 of 52 estimates (92%) were statistically non-significant, providing little evidence of consistent associations across studies and measurement approaches, thereby supporting the pooled estimates, which showed no evidence a robust association.

Page 9, lines 4-5 (Descriptive Results): "*None of the studies provided an estimate of the association between physical activity and TMS-derived measures in the published article.*"

Page 9, lines 28-31 (Descriptive Results): "*Of the 61 individual association estimates, 57 (93%) were not statistically significant. This included 26 of 27 (96%) CSE estimates, 14 of 15 (93%) SICI estimates, 9 of 9 (100%) ICF estimates, 5 of 5 (100%) SICF estimates, 2 of 2 (100%) cSP estimates, and 1 of 1 (100%) LICI estimate; both iSP estimates were statistically significant (0% were non-significant).*"

Response to Comment 8

Authors: We respectfully disagree that the absence of explicitly reported correlations in the published articles undermines the rationale for the meta-analysis. We believe it strengthens the contribution of our work, because it allowed us to recover and synthesize associations that had not previously been reported.

Meta-analysis is not limited to synthesizing effect estimates explicitly reported as primary outcomes. In this review, several studies provided sufficient statistical information in the published articles or open datasets, and additional association estimates were obtained directly from authors. These data allowed us to recover associations that had not been reported in the original publications and synthesize them across studies, thereby addressing an important question that had not previously been systematically examined.

We agree, however, that this rationale and contribution were not sufficiently clear in the original manuscript. We have therefore clarified in the Introduction that the relevant studies had collected both habitual physical activity and TMS-derived neurophysiological measures but had not directly reported their associations. We have also highlighted this as a strength of the study in the Discussion, emphasizing that the review involved secondary analyses of previously unreported association estimates, followed by their quantitative synthesis across studies.

Page 1, lines 24-26 (Abstract): "*We recovered previously unreported association estimates using statistical information from the articles, open datasets, and data requests to authors.*"

Page 3, line 17 (Eligibility Criteria): "*Studies were included […] irrespective of whether habitual physical activity was a primary exposure of interest*."

Page 13, lines 4-10 (Discussion): "*An important strength of this study is that among the articles that collected measures of both habitual physical activity and TMS-derived neurophysiology, none reported estimates of the associations between these variables. By recovering these estimates from reported statistical information, open datasets, and data requests to authors, we performed secondary analyses of previously unanalyzed data and synthesized the resulting associations across studies, generating new empirical evidence on a question that had not previously been examined.*"

Response to Comment 9
Thank you for spotting this. The sentence has been corrected and now reads: "*The multilevel random-effects meta-analysis showed no evidence of an association between physical activity and inhibitory processes*." (page 10, line 38)

Response to Comment 10
We agree that the evidence for cSP, iSP, and LICI is limited, with only one or two studies contributing to these measures. We have therefore clarified this point in the manuscript and have avoided interpreting the pooled estimates for cSP, iSP, and LICI as definitive evidence. However, we respectfully do not believe that these data should be removed from the manuscript, as they provide information about the current state of the evidence and were identified before the quantitative analyses were conducted.
Of note, if cSP, iSP, and LICI are not considered, SICI is the only remaining indicator of inhibitory processes. As shown in Figure 3, SICI was not associated with habitual physical activity ($r = -0.07$, 95% CI −0.17 to 0.04), a result similar to the pooled estimate including all four measures ($r = 0.02$, 95% CI −0.10 to 0.15). Although the pooled estimate changed modestly, both estimates were small and their confidence intervals included zero. Thus, excluding the sparsely represented measures would not alter the overall interpretation that there is no evidence of a robust association between habitual physical activity and inhibitory TMS-derived measures.

> Page 13, lines 21-28 (Discussion): "*For inhibitory measures, the pooled association was non-significant and practically negligible, with moderate heterogeneity and a wide prediction interval, indicating substantial uncertainty about the association in future studies. Subgroup analyses revealed differences across inhibitory measures, with a positive association observed for iSP only. However, the iSP estimate was based on only two effect sizes and should therefore be considered hypothesis-generating rather than evidence of a specific association between habitual physical activity and interhemispheric inhibition."*

Response to Comment 11
Thank you for spotting this. The sentence has been corrected and now reads:

> Page 13, line 39 (Discussion): "*Specifically, acute aerobic exercise produced a small increase in intracortical inhibition, reflected by a decrease in SICI (Hedges' g = 0.27),* […]"